NINE ADDITIONAL COUNTEREXAMPLES TO CONJECTURE 5 OF SONDOW, NICHOLSON, AND NOE

Abstract. Let R_n be the nth Ramanujan prime and put $\rho(n) = \pi(R_n)$. The $m \geq 20$ clause of Conjecture 5 of Sondow, Nicholson, and Noe asserts that $\rho(mn) \leq m\rho(n)$ for every $n \geq 2$. Axler later exhibited the exception $(m, n) = (38, 9)$ and asserted that it was unique. We give nine additional counterexamples and prove that, for $m \geq 20$, the complete exception set is

$$\{(m, 9) : m \in \{22, 23, 24, 25, 26, 32, 33, 36, 37, 38\}\}.$$

Consequently, the least valid threshold is $n = 10$ for exactly these ten values of m, and is $n = 2$ for every other $m \geq 20$.

All logarithms are natural. The nth Ramanujan prime is the least positive integer R_n such that

$$\pi(x) - \pi(x/2) \geq n \qquad (x \geq R_n),$$

and p_k denotes the kth prime. Sondow, Nicholson, and Noe [3, Conjecture 5] conjectured, in particular, that

$$\rho(mn) \leq m\rho(n) \qquad (m \geq 20,\ n \geq 2). \tag{1}$$

Axler subsequently asserted that the only exception to the full conjecture is $(38, 9)$ [1, Theorem 1.7]; the paper later appeared as [2].

Theorem 1. *For integers $m \geq 20$ and $n \geq 2$,*

$$\rho(mn) > m\rho(n) \quad \iff \quad n = 9 \ \text{ and } \ m \in E,$$

where

$$E = \{22, 23, 24, 25, 26, 32, 33, 36, 37, 38\}.$$

Equivalently, if

$$N_{\min}(m) = \min\{N \geq 2 : \rho(mn) \leq m\rho(n) \text{ for every } n \geq N\},$$

then

$$N_{\min}(m) = \begin{cases} 10, & m \in E, \\ 2, & m \geq 20 \text{ and } m \notin E. \end{cases}$$

Proof. We first verify the infinite tail. Put

$$a = 1.472, \qquad b = 2.51, \qquad c = \log 2, \qquad \phi(n) = a \log\log n + b.$$

2020 *Mathematics Subject Classification.* Primary 11N05; Secondary 11A41.

Key words and phrases. Ramanujan prime, prime-counting function, explicit estimate, computational number theory.

Axler's estimates [1, (5.5) and (5.9)] reduce the desired inequality, for $m \ge 20$ and $n \ge 1245$, to the positivity of a quantity $W_m(n)$ satisfying

$$\begin{aligned} W_m(n) >\ &\log m\left(c + \frac{c}{\log n} - \frac{\phi(n)}{\log^2 n} - \frac{\phi(n)}{\log^3 n}\right) - \frac{\phi(n)}{\log n} \\ &- \frac{\phi(n)(\log\log n - c)}{\log^2 n} \\ &+ \frac{(\log m + \frac{1}{2})c - 0.565}{\log(mn)} \\ &+ \frac{(c\log m - 0.565)\log\log n}{\log^2(mn)}. \end{aligned}$$

The last two terms are positive for $m \ge 20$. Define

$$q(n) = 0.03 + \frac{c}{\log n} - \frac{\phi(n)}{\log^2 n} - \frac{\phi(n)}{\log^3 n}.$$

Let $t_0 = \log 1245$, $\Phi(t) = a\log t + b$, and

$$Q(t) = 0.03t^3 + ct^2 - \Phi(t)(t+1).$$

Then $Q(t) = t^3 q(e^t)$ and

$$Q''(t) = 0.18t + 2c - \frac{a}{t} + \frac{a}{t^2} > 2.46 \qquad (t \ge t_0).$$

Moreover, $Q'(t_0) > 7.36$ and $Q(t_0) > 2.15$, so $q(n) > 0$ for $n \ge 1245$. These numerical inequalities follow, for example, from

$$7.126 < t_0 < 7.127, \qquad 0.6931 < c < 0.6932, \qquad 1.963 < \log t_0 < 1.964.$$

The functions

$$u_1(t) = \frac{\Phi(t)}{t}, \qquad u_2(t) = \frac{\Phi(t)(\log t - c)}{t^2}$$

are decreasing for $t \ge t_0$. Indeed, $u_1'(t) = (a - \Phi(t))/t^2 < 0$, while

$$t^3 u_2'(t) = H(t) := a(\log t - c) + \Phi(t) - 2\Phi(t)(\log t - c),$$

with

$$H'(t) = \frac{2(a - a(\log t - c) - \Phi(t))}{t} < 0 \quad \text{and} \quad H(t_0) < -6.44.$$

It follows that

$$\begin{aligned} W_m(n) &> (c - 0.03)\log 20 - u_1(t_0) - u_2(t_0) \\ &> 1.09. \end{aligned}$$

Thus

$$\rho(mn) \le m\rho(n) \qquad (m \ge 20,\ n \ge 1245). \tag{2}$$

It remains to consider $2 \le n \le 1244$. Define

$$\gamma(x) = \frac{c + c/\log x + 0.565/\log^2 x}{\log x + \log\log x - c - c/\log x}.$$

Axler's upper estimate [1, Theorem 1.2 and (3.4)–(3.5)] gives

$$\rho(k) < 2k(1+\gamma(k)) \qquad (k \geq 5225). \tag{3}$$

The numerator of γ is positive and decreasing, and its denominator is positive and increasing; hence γ is decreasing on this range. An exact computation gives

$$\min_{2\leq n\leq 1244}\left(\frac{\rho(n)}{2n}-1\right) = \frac{101}{1244}, \tag{4}$$

attained uniquely at $n = 1244$, where $R_{1244} = 24169$ and $\rho(1244) = 2690$. Also,

$$\gamma(5225) < \frac{0.694 + 0.694/8.56 + 0.565/8.56^2}{8.56 + 2.14 - 0.694 - 0.694/8.56} < 0.079 < \frac{101}{1244}.$$

Therefore, if $2 \leq n \leq 1244$ and $mn \geq 5225$, then

$$\rho(mn) < 2mn(1+\gamma(mn)) < m\rho(n).$$

Only

$$\mathcal{F} = \{(m,n) : m \geq 20,\ 2 \leq n \leq 1244,\ mn < 5225\}$$

remains. Since necessarily $n \leq 261$,

$$\#\mathcal{F} = \sum_{n=2}^{261}\left(\left\lfloor\frac{5224}{n}\right\rfloor - 19\right) = 21803.$$

For an integer $x \geq 0$, put

$$d(x) = \pi(x) - \pi(\lfloor x/2\rfloor).$$

By [3, Theorem 3], $R_j < p_{3j}$ for every j. Hence a sieve through

$$B = p_{15672} = 172069$$

contains every R_j with $j \leq 5224$. Since the function $\pi(x) - \pi(x/2)$ changes only at integer points,

$$R_j = 1 + \max\{0 \leq x < B : d(x) < j\}. \tag{5}$$

This is the last-deficit computation described in [3, Appendix].

Applying (5) and checking all pairs in $\mathcal{F}$ in integer arithmetic gives exactly the following failures. Here $R_9 = 71$ and $\rho(9) = 20$.

m	$9m$	R_{9m}	$\rho(9m)$	$20m$	excess
22	198	3167	448	440	8
23	207	3319	467	460	7
24	216	3467	486	480	6
25	225	3607	504	500	4
26	234	3761	523	520	3
32	288	4789	644	640	4
33	297	4967	664	660	4
36	324	5477	723	720	3
37	333	5651	743	740	3
38	342	5807	762	760	2

The last row is Axler's previously known exception; the first nine are additional. There are no other failures in $\mathcal{F}$. Together with (2) and (3), this proves the classification. The formula for $N_{\min}(m)$ follows immediately. $\square$

References

[1] C. Axler, *On the number of primes up to the nth Ramanujan prime*, arXiv:1711.04588v1 [math.NT], 2017.

[2] C. Axler, *On Ramanujan primes*, Funct. Approx. Comment. Math. **63** (2020), no. 1, 67–93, doi:10.7169/facm/1824.

[3] J. Sondow, J. W. Nicholson, and T. D. Noe, *Ramanujan primes: bounds, runs, twins, and gaps*, J. Integer Seq. **14** (2011), no. 6, Article 11.6.2, 11 pp.